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ORIGINAL ARTICLE

Evaluating the reliability, resilience, and vulnerability of water resources: the impact of minimum, maximum, environmental flows, and water quality

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Abstract

Surface water resources in the Gorganroud watershed are facing significant reliability challenges due to human activities and climate change. This research aims to assess the resilience, reliability, and vulnerability of water resources across 16 sub-watersheds, focusing on minimum, maximum and environmental flows as well as water quality. The findings indicate that the health of the minimum flow over periods of 7, 30, and 90 days is rated at 0.16, 0.32, and 0.45, respectively, reflecting unhealthy to moderate conditions at the watershed level. Additionally, the maximum flow health score of 0.34 highlights the necessity for effective flood management. Water quality assessments for drinking and agricultural purposes were estimated at 0.52 and 0.57, respectively; however, downstream agricultural sub-watersheds with intensive activities, such as Hajighoshan, Dasht, Gonbad, Ghazaghli, and Gorgan Dam, are facing critical conditions. The overall health of the environmental flow in the watershed is rated at 0.47, with the middle and lower sub-watersheds, including Dasht and Gorgan Dam, exhibiting poor conditions, while mountainous areas, such as Till Abad, demonstrate better performance. These results underscore the urgent need for integrated water resource management, vegetation restoration, and the control of agricultural pollution to enhance watershed health.

Keywords Watershed health · Environmental water requirement · RRV · Water quality · Hydrological approach

Introduction

A healthy watershed is vital for delivering essential ecosystem services, such as nutrient cycling, water purification, habitat protection, erosion and sediment control, flood regulation, climate moderation, natural hazard mitigation, and the provision of cultural resources (Hazbavi et al. 2018a, b; Alilou et al. 2019). Nonetheless, watersheds are increasingly subjected to pressures from human activities and environmental changes. To address these challenges effectively, it is imperative to implement integrated assessments that encompass socio-ecological perspectives and multidimensional strategies.

The assessment of watershed health encompasses the analysis of ecological, hydrological, and anthropogenic factors to identify deviations from natural conditions (Hazbavi, 2018; Chen 2022; Jiang 2024). River flow is a vital component for both ecosystems and human requirements. Consequently, monitoring changes in flow regimes is imperative for the conservation of biodiversity, as natural water flows are essential for supporting aquatic and riparian species (Yu 2022; Dhal and Kansal 2024). Exceeding critical thresholds in hydrological regimes can result in watershed degradation.

Key dimensions of watershed health include reliability, resilience, and vulnerability (Wu 2022). Resilience refers to the capacity of aquatic ecosystems to recover and meet human needs during stressful conditions. Research demonstrates that maintaining Environmental Water Requirements

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(EWR) strengthens both ecological and human water security (Jiao 2015; Yu et al. 2022). Conversely, vulnerability reflects the sensitivity of water resources to climatic, environmental, and anthropogenic pressures. Increasing land-use diversity and rising water demand have exacerbated watershed vulnerability on a global scale (Zhang et al. 2017).

Hydrological extremes, specifically minimum and maximum flows, serve as critical indicators in watershed studies. Minimum flows reflect water availability during drought conditions and are vital for effective water resource management (Morrison 2016). Conversely, maximum flows indicate flood potential and assess system stability during significant rainfall events. Research demonstrates that moderate floods can induce geomorphic changes, whereas extreme floods have the capacity to destabilize river channels (Leopold et al. 1964; Wolman and Miller 1960).

In addition to flow regimes, water quality is a critical factor in assessing watershed health. The physical, chemical, and biological properties of water significantly influence its usability and ecological functionality. Deteriorating water quality often results from agricultural runoff, land-use changes, and pollution, adversely affecting both human and ecological well-being (Chernos 2021). By integrating assessments of flow and water quality, we can obtain a comprehensive understanding of watershed reliability.

Previous studies have employed the Resilience, Recovery, and Vulnerability (RRV) framework to evaluate watershed health across various regions. Hazbavi and Sadeghi (2017) and Sadeghi et al. (2019a, b) identified a decline in resilience, increased vulnerability to drought, and a deficiency of healthy sub-watersheds in Shahroud. Sediq and Komori (2024) used RRV indices to assess water sustainability in the Kabul River Basin, drawing on CMIP6 GCM ensemble projections under SSP2-4.5 and SSP5-8.5 scenarios and SWAT simulations of surface runoff. Results indicate declining sustainability due to rising vulnerability and falling reliability/resilience, with future projections signaling heightened runoff risks in downstream, densely populated agricultural areas. This stresses the need for localized water management to counter climate uncertainties and ensure sustainable access. Zou et al. (2019) applied the RRV framework to assess adaptive measures under climate change scenarios, demonstrating that water conservation significantly reduces vulnerability, while dynamic flood control enhances reliability.

Rahim et al. (2021) integrated climatic indices with the RRV framework to evaluate freshwater reliability, revealing moderate yet vulnerable conditions. Chamani et al. (2023) investigated the Atrak watershed under future climate scenarios, emphasizing the need for adaptive management strategies.

More recently, Zeng et al. (2024) incorporated water allocation into the RRV framework, highlighting its crucial role in integrating water, food, and energy systems. Despite these valuable insights, a paucity of studies remains that simultaneously analyze hydrological variables, environmental flows, and water quality at the sub-watershed scale in climatically diverse regions.

Despite the increasing attention given to watershed health assessment in recent years, few studies have simultaneously considered both hydrological and water quality indicators within a unified analytical framework. This limitation has left a significant gap in understanding the integrated performance of river basins under human and climatic pressures. In this context, the present study seeks to address this gap by assessing the health of the Gorganroud watershed through the integration of minimum flow, maximum flow, environmental flow, and water quality indicators within the RRV framework. By doing so, this research provides a more comprehensive evaluation of watershed health and contributes to the development of more effective management strategies.

Materials and methods

The flowchart of the research methodology is shown in Fig. 1. The research process steps are explained as follows.

Study area

The Gorganroud watershed is located upstream of the Voshmgir Dam in Golestan Province, northeastern Iran. It encompasses an area of 7,166.87 km^2, with elevations ranging from 6 m in the lowlands to 2,887 m in the mountainous regions. The primary river within the watershed extends for 333 km. Mean annual precipitation varies between 231 and 848 mm, while average temperatures range from 11 to 18.1 °C, reflecting a notable climatic diversity that includes both humid and arid zones.

The land cover within the watershed demonstrates significant contrasts, with forests predominantly located in the southern and eastern regions, while agricultural and rangeland areas are prevalent in the northern and western sections. The watershed is confronted with a range of environmental challenges, including intense rainfall events, erosion-prone geology, and substantial deforestation attributed to dry farming practices. These factors contribute to increased runoff, soil erosion (Fig. 2), and sedimentation, thereby exacerbating watershed degradation (Azari et al. 2017).

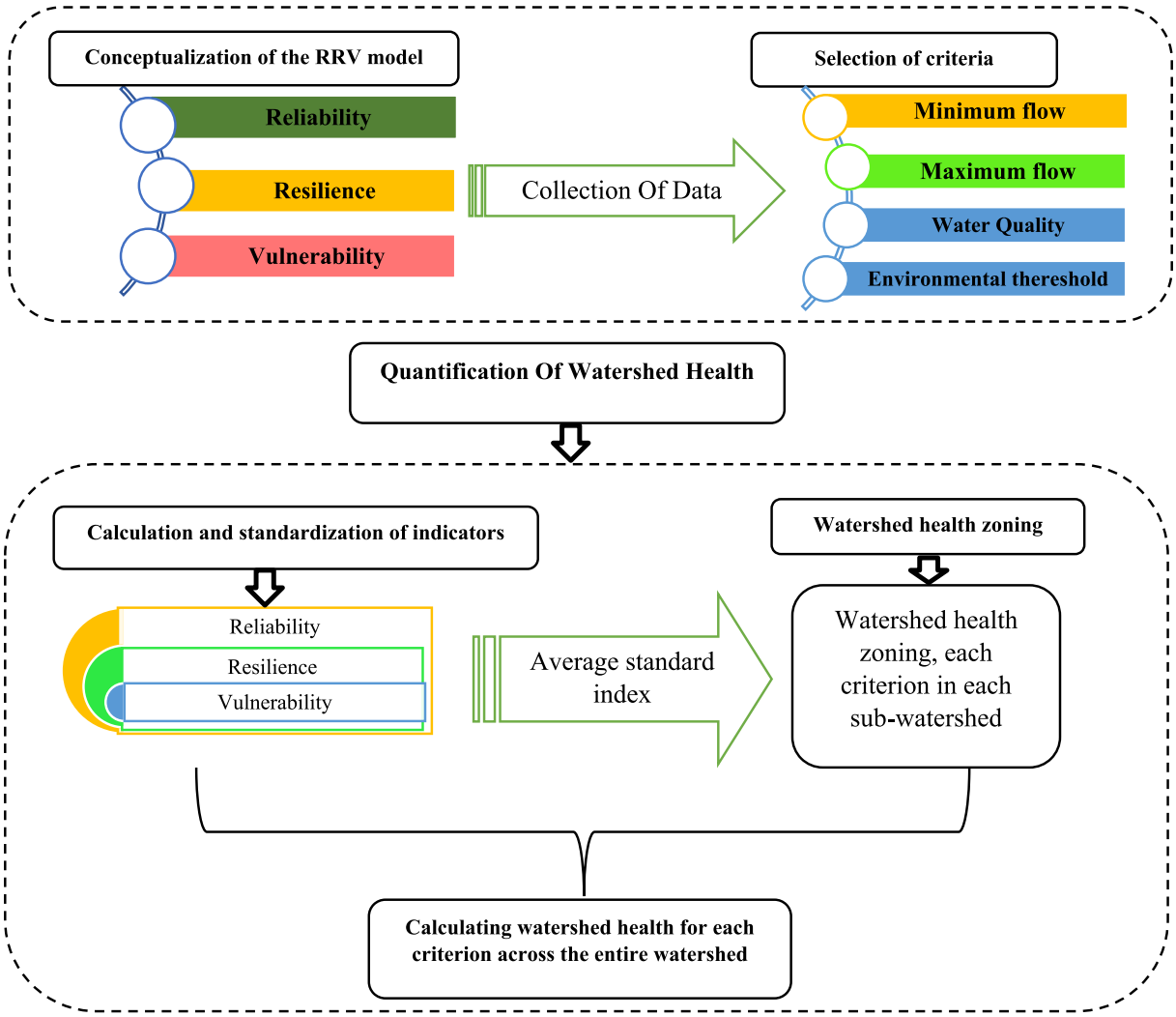

Fig. 1 Flow chart of the research methodology

The utilized datasets

To establish criteria for watershed health and implement the RRV framework, we gathered relevant data from all sub-watersheds. For each sub-watershed, we identified the hydrometric station with the most comprehensive and reliable records (refer to Table 1). The datasets comprised daily discharge measurements, station-specific characteristics, and surface water quality data from the sub-watershed outlets. To ensure comparability across the watershed, we utilized a consistent statistical period, employing a 20-year dataset from 2001 to 2020 for all sub-watersheds with hydrometric stations.

The impact of dams on RRV analysis was systematically addressed by incorporating storage capacity and regulated discharge records from the three primary dams within the Gorganroud watershed, as provided by the Iran Water Resources Management Company (IWRMC). These datasets were utilized to enhance flow estimations—specifically for peak and environmental discharges—thereby reflecting the regulatory effects of dam operations on the hydrological regime and, subsequently, on the ecological integrity of the sub-watershed.

The Gorganroud watershed in northeastern Iran is a vital area for water resource studies, owing to its hydrological significance and climatic diversity. Sixteen key hydrometric stations, as detailed in Table 1, play a crucial role in monitoring both discharge and water quality. These stations, numbered S1 to S16, are strategically positioned across various sub-watersheds to ensure comprehensive coverage of the watershed's hydrological characteristics and land use. For example, stations located near the watershed outlet, such as S16, are influenced by extensive agricultural practices and water harvesting, while upstream stations like S7 and S13 benefit from forest cover. These stations enable the analysis of flow dynamics and water quality and have been

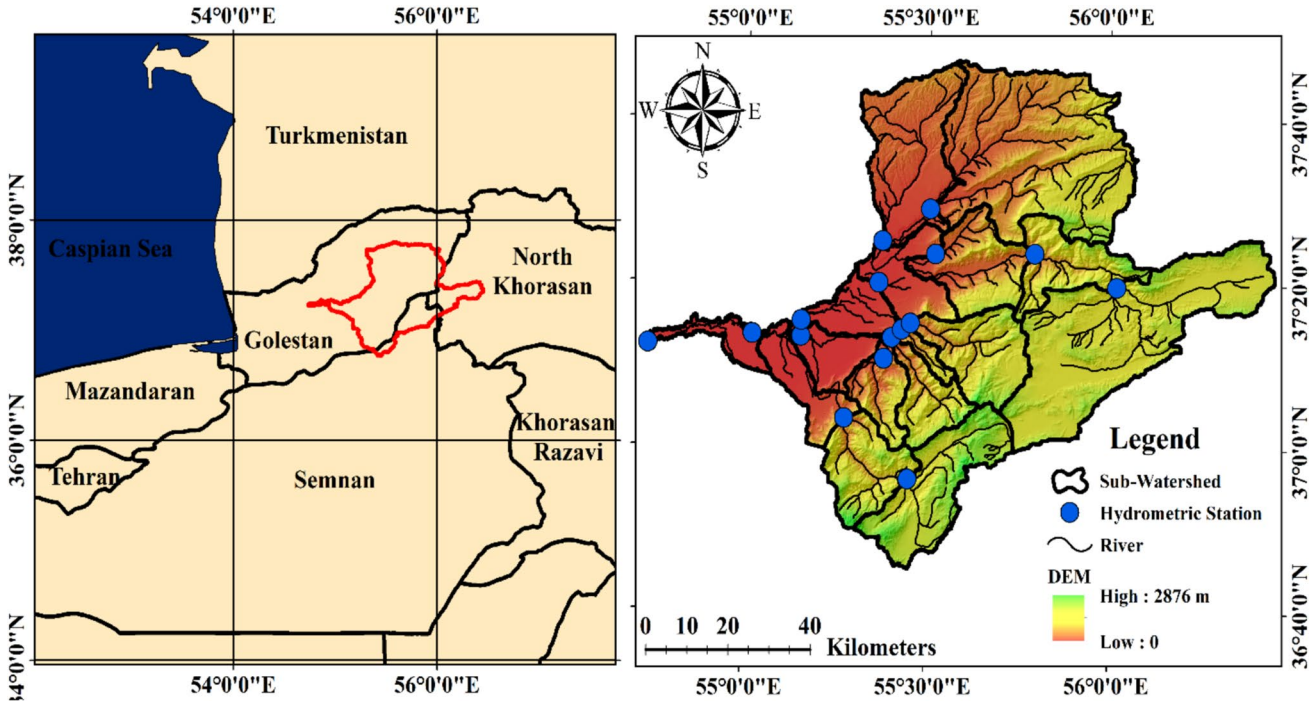

Fig. 2 Geographical location of the Gorganroud watershed in Iran

Table 1 Geographical characteristics of hydrometric stations

Row	Sub-watershed	Station	Elevation above sea level (m)	Geographical latitude (degrees)	Geographical longitude (degrees)
1	S1	Hajighoshan	73	37.4	55.4
2	S2	Tamar	115	37.5	55.5
3	S3	Tangrah	459	37.4	55.8
4	S4	Gharehshor	77	37.3	55.4
5	S5	Ghochmaz	181	37.4	55.5
6	S6	Dasht	1010	37.3	56
7	S7	Galikesh	237	37.3	55.5
8	S8	Tillabad	912	36.9	55.5
9	S9	Gholitapeh	243	37.2	55.4
10	S10	Lazoreh	196	37.2	55.4
11	S11	Nodeh	280	37.1	55.3
12	S12	Gonbad	37	37.3	55.2
13	S13	Jangaldeh	180	37.2	55.4
14	S14	Arazkoseh	35	37.2	55.2
15	S15	Ghazaghli	25	37.2	55
16	S16	Gorgan Dam	2	37.2	

employed to assess the resilience, reliability, and vulnerability of water resources using the RRV framework. The data collected from these stations indicate significant disparities in the hydrological health of the sub-watersheds, providing a robust scientific foundation for the integrated management of water resources within the Gorganroud watershed.

Models and tools used

RRV approach

The Reliability–Resilience–Vulnerability (RRV) framework has become a strong and comprehensive tool for assessing

watershed health, presenting several advantages over traditional models and methodologies. Its strengths come from its inherent qualities, straightforward methodology, and alignment with scientific research and management goals. Unlike many existing approaches, the RRV framework simultaneously evaluates three vital aspects of system performance—reliability, resilience, and vulnerability—offering an integrated, multidimensional view of watershed conditions.

A key strength of this framework is its effectiveness in decision-making; the results, presented in both numerical and classified forms, facilitate the identification of critical weaknesses and the formulation of practical, evidence-based management strategies. In this context, reliability (Rel) is defined as the frequency of system failures (Silva, 2011), where a failure refers to the inability of a watershed to operate within an acceptable performance range (Lu et al. 2015). The validity and applicability of the RRV framework have been thoroughly validated across diverse geographical and hydrological contexts, as evidenced by a substantial body of literature (e.g., Hashimoto et al. 1982; Hazbavi et al. 2018a, b; Sadeghi et al. 2019a, b). Both global and local studies underscore its effectiveness in enhancing watershed assessment and management.

However, before calculating the $R_{el}R_{es}V_{ul}$ indices, it is essential to normalize the dataset to eliminate size effects and ensure meaningful comparability across different watersheds.

For this purpose, Eq. (1) is used:

$$Y = \frac{x_i - x_{min}}{x_{max} - x_{min}} \tag{1}$$

where Y is the standardized value, xi is the observed value, and x_{max} and x_{min} are the maximum and minimum observed values, respectively.

Reliability (R_{el}) is calculated using Eq. (2) as follows:

$$Reliability(R_{el}) = 1 - \frac{\sum_{j=1}^{M} d[j]}{T} \tag{2}$$

where M is the number of failure events, $d[j]$ is the duration of the jth failure event, and T is the total number of time intervals (Hashimoto et al. 1982; Silva and De 2010; Hoque et al. 2012).

$$Resilience(R_{es}) = \left\{ \frac{1}{M} \sum_{j=1}^{M} d[j] \right\}^{-1} \tag{3}$$

R_{es} is the inverse of the mean time between failures. Therefore, the longer the average duration of the failure sequence, the more difficult it is for a watershed to recover from failure (Hashimoto et al. 1982; Silva 2010; Hoque et al. 2012).

Vulnerability is defined as the average maximum failure in each sequence of continuous failures. Given these conditions and according to Eq. (4), the vulnerability equation (V_{ul}) becomes:

$$Vulnerability(V_{ul}) = \frac{1}{M} \sum_{j=1}^{T} \left\{ \left[\frac{L_{obs}(i) - L_{std}(i)}{L_{std}(i) - L_{min}(i)} \right] H\left[L_{obs} - L_{std}(i)\right] \right\} \tag{4}$$

where $L_{obs}(i)$ is the observed study constituent at the ith time step, $L_{std}(i)$ is the corresponding compliance standard, and $H[]$ is the Heaviside function, which ensures that only failure events are involved in the V_{ul} calculation in Eq. (4). The Heaviside function is a mathematical and discontinuous function whose value is zero for negative arguments and one for positive arguments. Therefore, the value of V_{ul} is negative and contrary to the two indices R_{el} andR_{es}, and to create consistency between the three indices, this index is adjusted as follows, based on Eq. (5) (Ervinia et al. 2019).

$$V = 1 - V_{vul} \tag{5}$$

Calculation of health indicators

Considering that after determining the $R_{el}R_{es}V_{ul}$ indices and specifying each of them for the desired period in the watershed area, it is necessary to present and display the overall health status of the watershed, the average of the $R_{el}R_{es}V_{ul}$ indices are taken to incorporate the impact of each index on the watershed's health and provide an overall view of the watershed's health (Sadeghi et al. 2019a, b). To calculate the final RelResVul, the geometric mean of the three indices of reliability, resilience, and vulnerability is taken, considering the necessity of greater sensitivity to changes in individual variables based on the Eq. (6) (Ervinia et al. 2019).

$$(R_{el}R_{es}V_{ul}) = \left[\prod_{i=1}^{3} R_{el}.R_{es}.V_{ul} \right]^{\frac{1}{3}} \tag{6}$$

Watershed health classification

To describe and interpret the health of watersheds and their conditions (Table 2), the watershed health indicator is categorized into five groups (Sadeghi and Hazbavi 2018).

Table 2 Classification of watershed health status

Range	0–0.2	0.21–0.4	0.41–0.6	0.61–0.8	0.81–1
Group	Unhealthy	Relatively unhealthy	Medium	Healthy	Very healthy

Calculation criteria in watershed health evaluation

Minimum flows

The Indicators of Hydrologic Alteration (IHA) software was utilized to quantify hydrological flow metrics. Developed by the Natural Science and Technology Center (NSTC) at the Nature Conservancy in Denver, Colorado, this tool provides a comprehensive framework for analyzing hydrological regimes. The IHA framework encompasses a range of indices—magnitude, duration, timing, frequency, and rate of change—that are critical for evaluating flow conditions. The IHA has been extensively applied across diverse geographic and climatic contexts, demonstrating strong reproducibility and reliability, as evidenced by numerous studies (e.g., Chen et al. 2023; Nasiri Khiavi et al. 2023).

This study estimated the base flow index and critical thresholds for minimum discharges over 7-, 30-, and 90-day intervals at the outlets of the investigated sub-watersheds. To accomplish this, daily discharge records from 16 hydrometric stations were analyzed, covering the period from 2001 to 2020, with the specific timeframe determined by data availability at each station. The records were processed using IHA software to derive the relevant indices, which were subsequently subjected to statistical analyses for each station.

Flow duration curves (FDCs) were subsequently developed to evaluate the probability of minimum flows over durations of 7, 30, and 90 days. The FDC functions as both a graphical and analytical instrument, depicting the probability distribution of flows over time by ranking discharge values from highest to lowest and calculating the percentage of time each discharge is equaled or exceeded. This approach facilitates precise estimation of flow thresholds and establishes a robust framework for analyzing low-flow conditions.

Maximum flow

The stage-discharge relationship quantifies the correlation between river stage (water level) and discharge at a specific cross-section, typically represented as a curve or by Eq. (7). This relationship facilitates the estimation of discharge based on stage observations and is widely employed in hydrometric stations to monitor river flow across various hydrological conditions, ranging from low flows to flood peaks. As water levels rise, corresponding increases in discharge can be calculated, allowing for the estimation of instantaneous discharge, particularly peak discharge, using the stage-discharge curve. Instantaneous peak discharge, defined as the maximum flow recorded during a flood event or within a specific time interval, represents the highest rate of water flow through a river section at any given moment. This parameter is critical for the design and safety assessment of hydraulic structures, such as dams, bridges, and drainage systems, as it provides essential information for flood risk evaluation and hydraulic capacity analysis.

In the present study, the reliability of maximum discharge in the Gorganroud sub-watersheds was evaluated using instantaneous peak discharge data sourced from the Golestan Regional Water Company (GRWC). Discharge thresholds for each station were established based on the stage–discharge relationship (Eq. (7)). Subsequently, the assessment of maximum discharge health in each sub-watershed was conducted utilizing the RRV approach.

$$Q = a(H - H_0)^b \tag{7}$$

where Q is the flow rate (m^3/s), H the water height or stage level at the specified section (m), H_0 the zero scale or reference level (m), and a and b are experimental coefficients determined based on the measured data for each river section.

Environmental flow estimation

The environmental flow framework is crucial for understanding the ecological and hydrological dynamics of aquatic systems. Developed by. Tennant (1976), the Tennant method offers a systematic approach to determining the minimum flow required to maintain the health and biological integrity of aquatic ecosystems. By analyzing data from 58 cross-sections across 11 rivers in Montana, Nebraska, and Wyoming, Tennant established that a minimum flow of 10% of the mean annual flow (MAF) is necessary for the short-term survival of fish, 30% MAF supports relatively good survival, and 60% MAF is essential for providing suitable habitat.

Although the Tennant method is widely utilized and practical, it has certain limitations, particularly its reliance on the Mean Annual Flow (MAF), which may not adequately account for seasonal variations or the specific ecological requirements of different aquatic species (Raouf and Ali Oghlu 2020). In this study, we determined the threshold for minimum river flow to be 30% of the MAF for each monitoring station.

Water quality

Measuring water quality variables for drinking, agriculture, and industry is crucial to ensure compliance with established standards. To evaluate the health of water quality across the drinking and agricultural sectors within each sub-watershed of the Gorganrud watershed, water quality data were collected from the Gorganrud Watershed Council (GRWC). Health thresholds for each variable were established in accordance with national standards set by the Iranian Research Institute for drinking, agricultural, and industrial applications. Ultimately, the overall health of water quality

in each sub-watershed was assessed using the Relative Risk Value (RRV) model (Zare Gharizi et al. 2024).

Drinking water quality

The standard for key variables of drinking water that focus on human health is mentioned in Table 3.

Agricultural water quality

In agriculture, water quality is evaluated based on its impact on plant growth; therefore, the variables affecting the suitability of water quality for agriculture are listed in Table 4.

Calculation of the health index

The final health index is calculated based on two components: "$RRV_{drinkingwaterquality}$" and "$RRV_{agriculturalwaterquality}$." The reason for selecting these two indicators is their central and ultimate role in reflecting the ecological and human health status of the sub-watershed. Water quality indicators represent the final outcome of the interaction of various factors such as flow fluctuations, land use, surface runoff, chemical fertilizers, human activities, and climatic conditions. These two indices not only indicate the status of direct water resource utilization but also indirectly reflect the performance of other hydrological components. The use of these two indicators follows the approach of Hazbavi et al. (2018a, b) and Hoque et al. (2012) is consistent, suggesting that in watershed health analyses, water quality should be considered as a representative of hydrological, human, and ecological impacts. In Eq. (8) with the aggregation method, an equal weighted sum method ($W_1 = W_2 = 0.5$) has been used, as both criteria (drinking and agriculture) have similar weights and balanced importance in terms of performance and management in the area. As Ervinia et al. (2019) have pointed out, the incremental aggregation method is appropriate when the indicators have complementary roles and similar importance and are linked to managerial decision-making objectives.

$$RRV_{drinkingwaterquality} \times W_1 + RRV_{agriculturalwaterquality} \times W_2 = HealthIndex \quad (8)$$

Results

Results of minimum flows health

After establishing the hydrological criteria, we calculated the resilience, reliability, and vulnerability indices at the sub-watershed scale for minimum flows of 30, 7, and 90 days. The health zoning map (Fig. 3) reveals that the minimum 7-day flow values are generally low, indicating that most sub-watersheds are vulnerable in the short term. Specifically, sub-watersheds S14 and S15 are the most at risk, while S11 is the least vulnerable. In terms of the minimum 30-day flow values, there has been an average improvement, ranging from 0.14 in S12 to 1.0 in S4. Sub-watershed S4 stands out with the highest health value, reflecting strong resilience during average conditions. Furthermore, the health values for the 90-day flow are generally higher than those for the 7 and 30-day minimum flows, ranging from 0.2 in S6 to 1.0 in S4, indicating greater resilience of the watersheds over longer periods.

Table 3 Variables for determining the health of drinking water

Variable	Na^+	Mg^{2+}	Ca^{2+}	HCO_3^-	SO_4^{2-}	Cl^-	Total dissolved solid (TDS)	pH	Total hardness (TH)
Maximum allowable concentration	250	–	–	–	400	400	1500	5–6.9	500
Maximum desired concentration	200	30	300	200	250	250	1000	5.5–6.8	200

Table 4 Variables for determining the health of the agricultural water

Variable	Electrical conductivity (EC)	Sodium adsorption ratio (SAR)	Cl^-	B
Maximum allowable concentration	X < o.7	X < 10	Low concentrations	Low concentrations

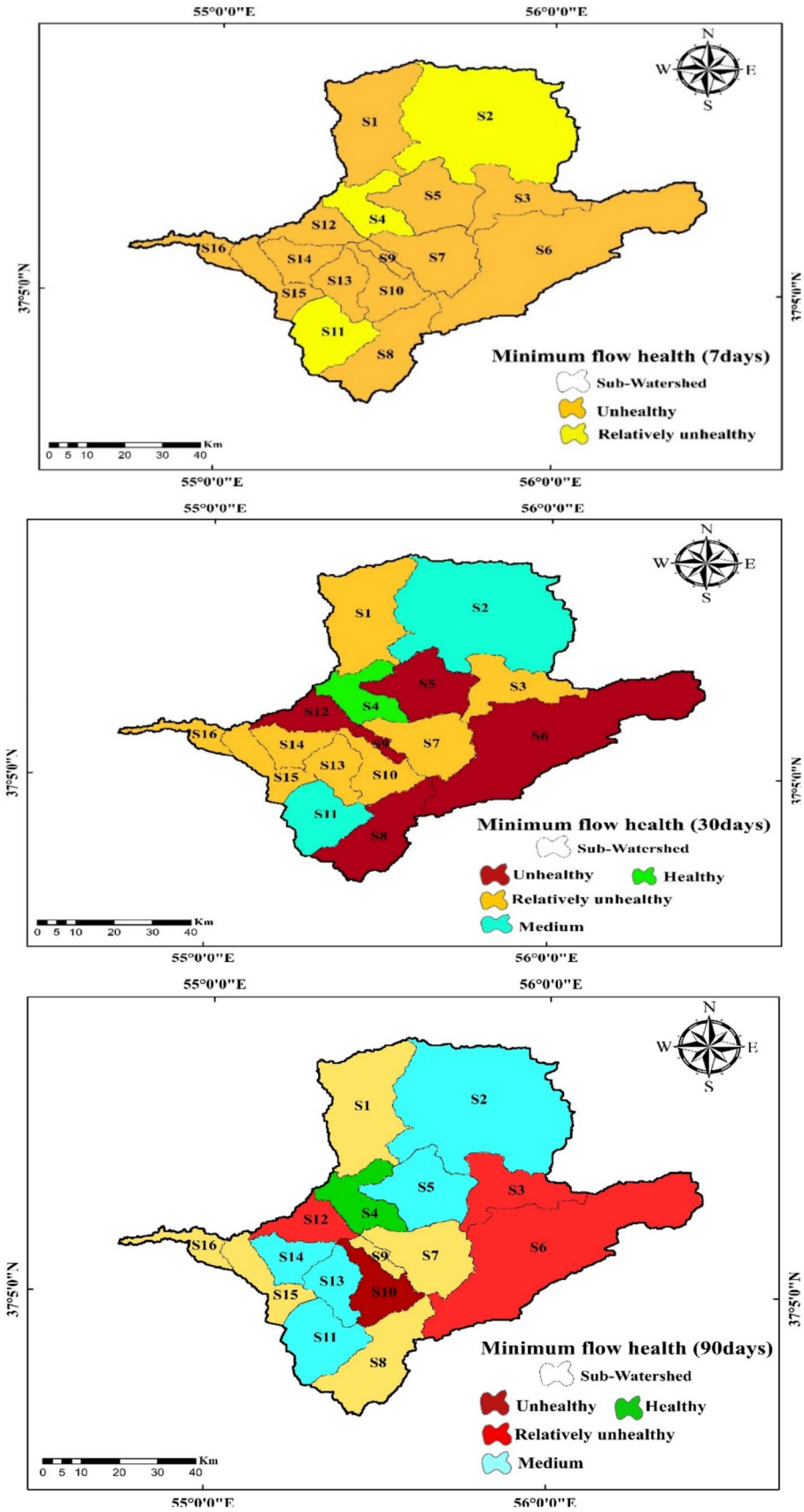

Fig. 3 Sub-watersheds' health of the Gorganrud watershed based on minimum flows (7, 30, and 90 days)

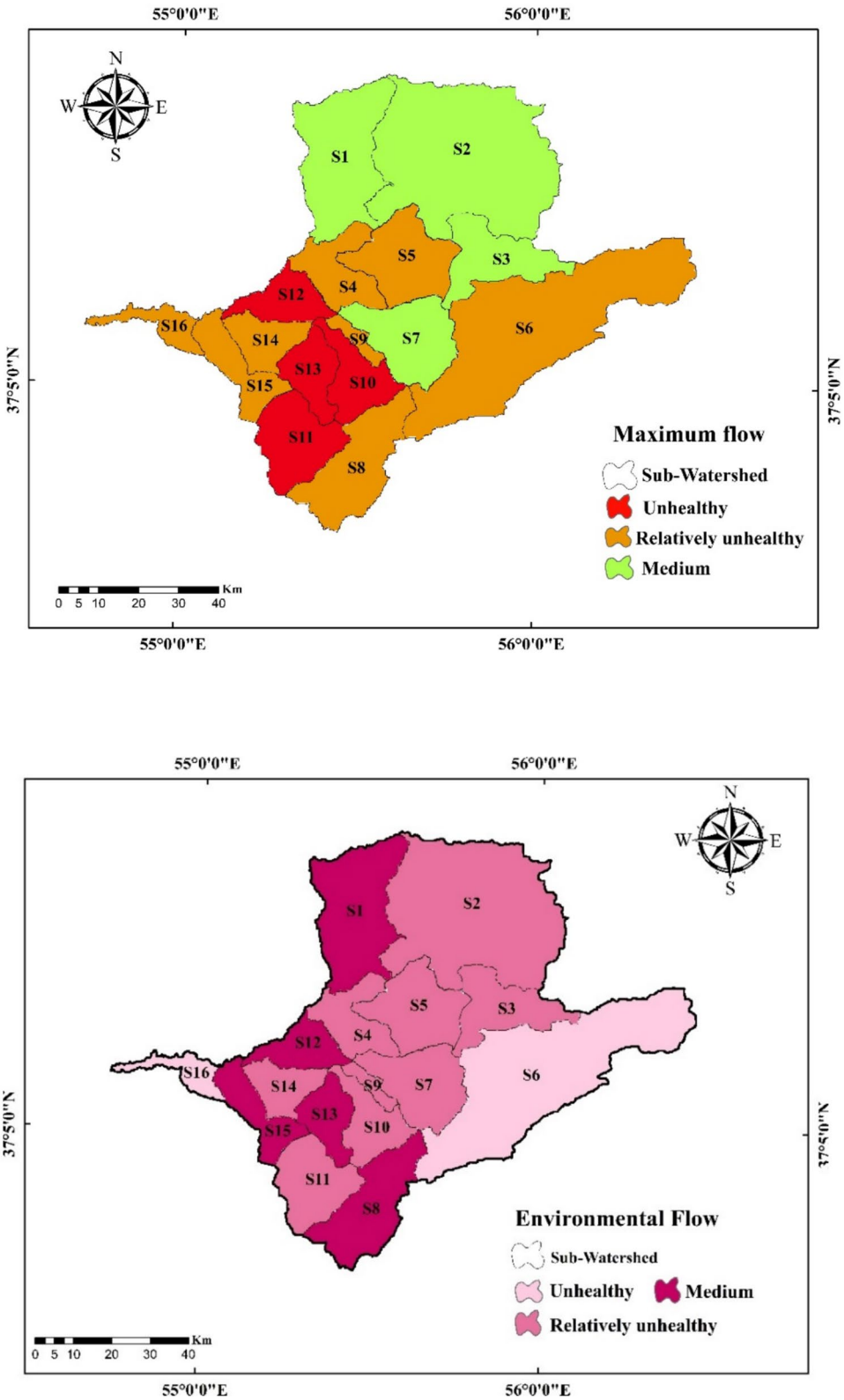

Fig. 4 Sub-watershed health of the Gorganrud watershed based on maximum flow

Results of maximum flow health

The maximum flow discharge health zoning map (Fig. 4) for the Gorganroud watershed illustrates health values across the sub-watersheds, ranging from 0.06 (S13) to 0.55 (S7). These values indicate the sub-watersheds' resilience and vulnerability to flooding, with S13 identified as the most vulnerable and S7 as the most resilient. Sub-watersheds S7, S3, and S9, which possess resilience ratings exceeding 0.5 (RRV > 0.5), demonstrate high resilience and stability alongside low vulnerability. Those with moderate resilience, stability, and vulnerability, ranging from 0.3 to 0.5 (0.5 ≥ RRV ≥ 0.3), include S1, S2, S5, S6, S8, S11, S12, S14, S15, and S16. Finally, sub-watersheds S4, S10, and S13 show low resilience, low stability, and significant vulnerability (RRV < 0.3).

Results of environmental flow health

According to Fig. 5, the values of environmental flow health vary in the range of 0.31 to 0.66, with the lowest and highest health being in sub-watersheds S6 and S15, respectively.

Results of water quality

This study evaluated two criteria—water use for drinking and agriculture—in relation to water quality health. After calculating the variable values (refer to Table 5) pertinent to the health of drinking and agricultural water quality, we assessed these criteria at both the sub-watershed and watershed levels. The results indicated that in the sensitive sub-watersheds, several variables raised concerns: water hardness was very high in sub-watersheds S1, S15, and S16. Additionally, sub-watersheds S1, S6, S12, S15, and S16 exhibited elevated salinity levels, as indicated by Electrical Conductivity (EC), rendering them unsuitable for agricultural purposes. Furthermore, the Sodium Absorption Ratio (SAR) was significantly high in sub-watersheds S1, S12, and S16, which increases the risk of soil salinity.

The results of Table 6 show the analysis of the concentration of anions, the concentration of cations, and the type of water for each sampling station, and also Fig. 6 shows the chemical analysis of the elements in each sub-watershed. The concentration of anions in the water samples indicates that in the sub-watersheds S1, S2, S3, S4, S6, S15, and S16, the element (Cl^-) is dominant as an anion, and the order of the elements is as follows ($Cl^- > SO_4^{2-} > HCO_3^-$). In the sub-watersheds S5, S6, S11, S12, S15, and S16, chloride (Cl) is dominant as an anion, and the order of elements is as follows ($Cl^- > HCO_3^- > SO_4^{2-}$). The S4, S5, S6, S11, S12, S13 and S14 bicarbonate (HCO_3^-) sub-watersheds are dominant as anion ($HCO_3^- > SO_4^{2-} > Cl^-$).

The predominance of chlorine or bicarbonate in water samples can be attributed to geological facies, land use, and potential sources of pollution in their respective sub-watersheds. The concentration of cations in the water samples shows that in the sub-watersheds S1, S2, S3, S6, S15, and S16, the dominant cations are sodium (Na^+) and potassium (K^+), and the order of the elements is

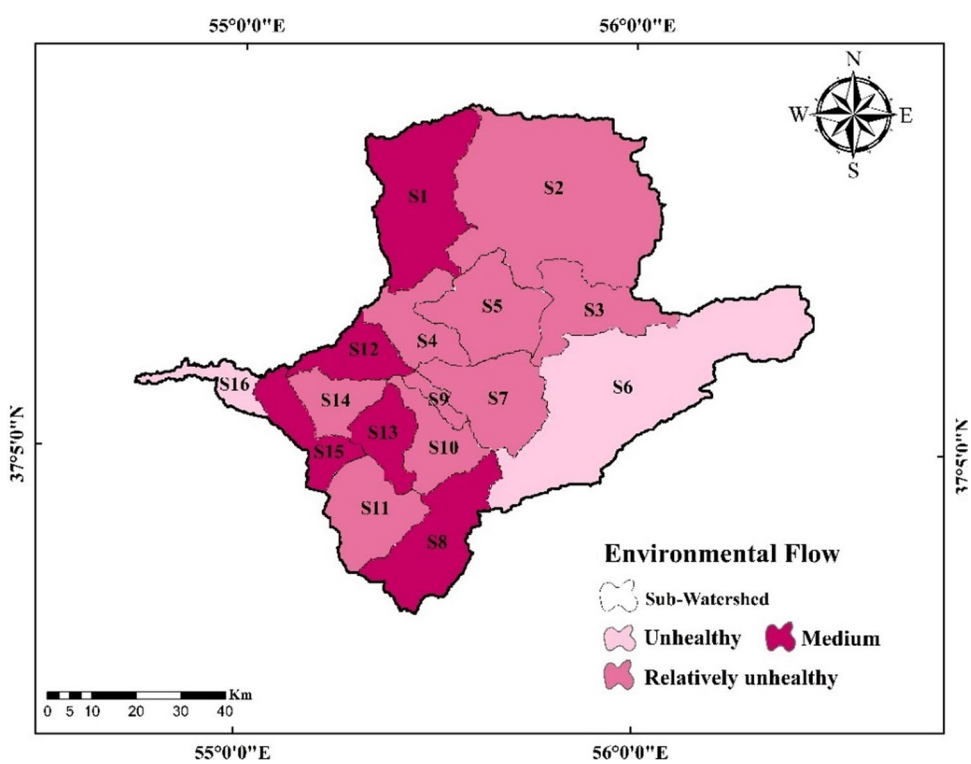

Fig. 5 Sub-watershed health of the Gorganrud watershed based on the environmental flow

Table 5 The values of drinking and agricultural water quality variables in each sub-watershed

Row	Sub-watershed	Station	Total Hardness (TH) mg/L	Sodium adsorption ratio (SAR) meq/L	Electrical conductivity (EC) (μS/cm)	Na%	Total cation	Total anion	Ca^{2+} (mg/l)	pH
1	S1	Hajighoshan	1727.43	9.56	6949.31	53.51	74.91	74.67	278.27	7.84
2	S2	Tamar	630.23	0.82	1837.63	24.47	16.79	21.21	142.09	7.80
3	S3	Tangrah	271.21	2.08	858.58	38.92	8.94	8.36	53.08	7.79
4	S4	Gharehshor	335.18	1.50	922.92	29.38	9.57	8.28	57.10	7.83
5	S5	Ghochmaz	267.49	0.86	649.34	21.2	6.84	6.30	42.67	7.91
6	S6	Dasht	1225.38	4.76	2778.14	40.62	41.66	39.63	137.69	11.75
7	S7	Galikesh	220.23	0.70	517.32	19.66	5.52	4.88	43.62	7.88
8	S8	Tillabad	295.20	5.26	1508.63	60.6	15.09	15.37	56.61	7.90
9	S9	Gholitapeh	215.23	0.35	462.09	19.39	5.38	4.86	45.63	7.84
10	S10	Lazoreh	243.99	1.42	436.69	31.52	7.17	6.91	51.48	8.12
11	S11	Nodeh	301.61	5.44	1518.51	61.06	15.60	15.58	57.61	7.87
12	S12	Gonbad	1128.29	7.09	4412.40	51.33	46.75	45.45	173.34	7.80
13	S13	Jangaldeh	222.28	0.66	525.02	18.95	5.52	5.48	46.37	7.82
14	S14	Arazkoseh	416.40	3.82	1559.62	48.54	16.31	16.34	73.78	7.84
15	S15	Ghazaghli	764.34	5.89	3033.69	51.63	31.85	31.01	123.26	7.82
16	S16	Gorgan dam	889.96	6.91	3626.59	53.72	38.77	38.87	141.22	7.78

Table 6 Analysis of the concentration of anions and cations in each sub-watershed

Row	Sub-watershed	Station	Concentration of anions	Concentration of cations	Water type
1	S1	Hajighoshan	$Cl^- > SO_4^{2-} > HCO_3^-$	$Na^+{}_+K^+ > Mg^{2+} > Ca^{2+}$	Cl^-
2	S2	Tamar	$Cl^- > SO_4^{2-} > HCO_3^-$	$Ca^{2+} > Mg^{2+} > Na^+{}_+K^+$	Cl^-
3	S3	Tangrah	$Cl^- > HCO_3^- > SO_4^{2-}$	$Na^+{}_+K^+ > Mg^{2+} > Ca^{2+}$	Cl^-
4	S4	Gharehshor	$HCO_3^- > SO_4^{2-} > Cl^-$	$Mg^{2+} > Ca^{2+} > Na^+{}_+K^+$	HCO_3^-
5	S5	Ghochmaz	$HCO_3^- > Cl- > HCO_3^-$	$Mg^{2+} > Ca^{2+} > Na^+{}_+K^+$	HCO_3^-
6	S6	Dasht	$Cl > SO_4^{2-} > HCO3^-$	$Mg^{2+} > Na^+{}_+K^+ > Ca^{2+}$	Cl^-
7	S7	Galikesh	$HCO_3^- > Cl^- > SO_4^{2-}$	$Mg^{2+} > Ca^{2+} > Na^+{}_+K^+$	HCO_3^-
8	S8	Tillabad	$Cl^- > HCO_3^- > SO_4^{2-}$	$Na^+{}_+K^+ > Mg^{2+} > Ca^{2+}$	Cl^-
9	S9	Gholitapeh	$HCO_3^- > SO_4^{2-} > Cl^-$	$Ca^{2+} > Mg^{2+} > Na^+{}_+K^+$	HCO_3^-
10	S10	Lazoreh	$HCO_3^- > Cl^- > SO_4^{2-}$	$Ca^{2+} > Mg^{2+} > Na^+{}_+K^+$	HCO_3^-
11	S11	Nodeh	$Cl^- > HCO_3^- > SO_4^{2-}$	$Na^+{}_+K^+ > Mg^{2+} > Ca^{2+}$	Cl^-
12	S12	Gonbad	$Cl^- > SO_4^{2-} > HCO_3^-$	$Na^+{}_+K^+ > Mg^{2+} > Ca^{2+}$	Cl^-
13	S13	Jangaldeh	$HCO_3^- > SO_4^{2-} > Cl^-$	$Ca^{2+} > Mg^{2+} > Na^+{}_+K^+$	HCO_3^-
14	S14	Arazkoseh	$Cl^- > SO_4^{2-} > HCO_3^-$	$Na^+{}_+K^+ > Mg^{2+} > Ca^{2+}$	Cl^-
15	S15	Ghazaghli	$Cl^- > SO_4^{2-} > HCO_3^-$	$Na^+{}_+K^+ > Mg^{2+} > Ca^{2+}$	Cl^-
16	S16	Gorgan Dam	$Cl^- > SO_4^{2-} > HCO_3^-$	$Na^+{}_+K^+ > Mg^{2+} > Ca^{2+}$	Cl^-

$Na^+ + K^+ > Mg^{2+} > Ca^{2+}$. In the sub-watersheds S4, S5, S7, S11, S12, and S13, magnesium (Mg^{2+}) and calcium (Ca^{2+}) are the dominant cations ($Mg > Ca^{2+} > Na^+ + K^+$).

The sub-watersheds S8, S9, S10, S13, and S14 are the dominant cations of calcium (Ca^{2+}) and magnesium (Mg^{2+}) ($Ca^{2+} > Mg^{2+} > Na^+ + K^+$Sodium–Potassium or Calcium-Magnesium Cations). These elements are usually naturally present in river, sea, and lake waters and can enter surface and groundwater under the influence of erosion of rocks and soil. Based on the obtained results, water samples in the Gorganroud watershed can be classified into two main types of water:

1."Cl-" type: sub-watersheds S1, S2, S3, S6, S10, S15, and S16 are classified as chloride-type waters, characterized by chloride as the dominant anion.

2."HCO_3^-" type: sub-watersheds S4, S5, S7, S11, S12, S13, and S14 fall into the bicarbonate water type group, characterized by "HCO_3^-" as the dominant anion. The type of water can provide insights into geochemical processes, water–rock interactions, and potential pollution sources in the sub-watershed.

Results of the drinking and agricultural water quality health

The health values of drinking water (Fig. 7) range from 0.28 to 0.88. Among the sub-watersheds, S13 (0.88) and S14 (0.72) demonstrate the highest health values, indicating superior drinking water quality. Conversely, S1 (0.28), S6 (0.37), S15 (0.32), and S16 (0.37) exhibit the lowest health values, suggesting inferior drinking water quality. The remaining sub-watersheds have health values ranging from 0.34 to 0.69, reflecting average water quality.

Discussion

Analysis of minimum flow health

Figure 3 illustrates that the majority of sub-watersheds exhibit an increase in health values from RRV7 to RRV90, indicating improved resilience over time. Notably, sub-watersheds S4, S11, and S2 demonstrate higher minimum flow health across 7, 30, and 90-day periods, underscoring their enhanced resilience. The superior performance of S2 and S4 can be attributed to S2's high flow regime and S4's central hydrological position, both of which contribute to stable water availability.

In S11, health values of 0.30 (RRV7), 0.52 (RRV30), and 0.56 (RRV90) suggest adequate short-term resilience, with a notable improvement over longer durations. In contrast, S8 shows relatively weak resilience at short timescales. Although it has a dense drainage network, minimum flow health improves significantly at 30 and 90 days, indicating increased resilience over longer periods but vulnerability in the short term.

Sub-watershed S13, characterized by extensive forest cover, consistently exhibits favorable minimum flow health across all temporal scales. This finding aligns with Lin et al. (2021), who highlighted the critical role of vegetation in maintaining stable streamflows. In contrast, S16, located at the watershed outlet, shows poor short-term resilience, evidenced by notably low 7-day minimum flow health. This vulnerability is linked to intensive surface and groundwater

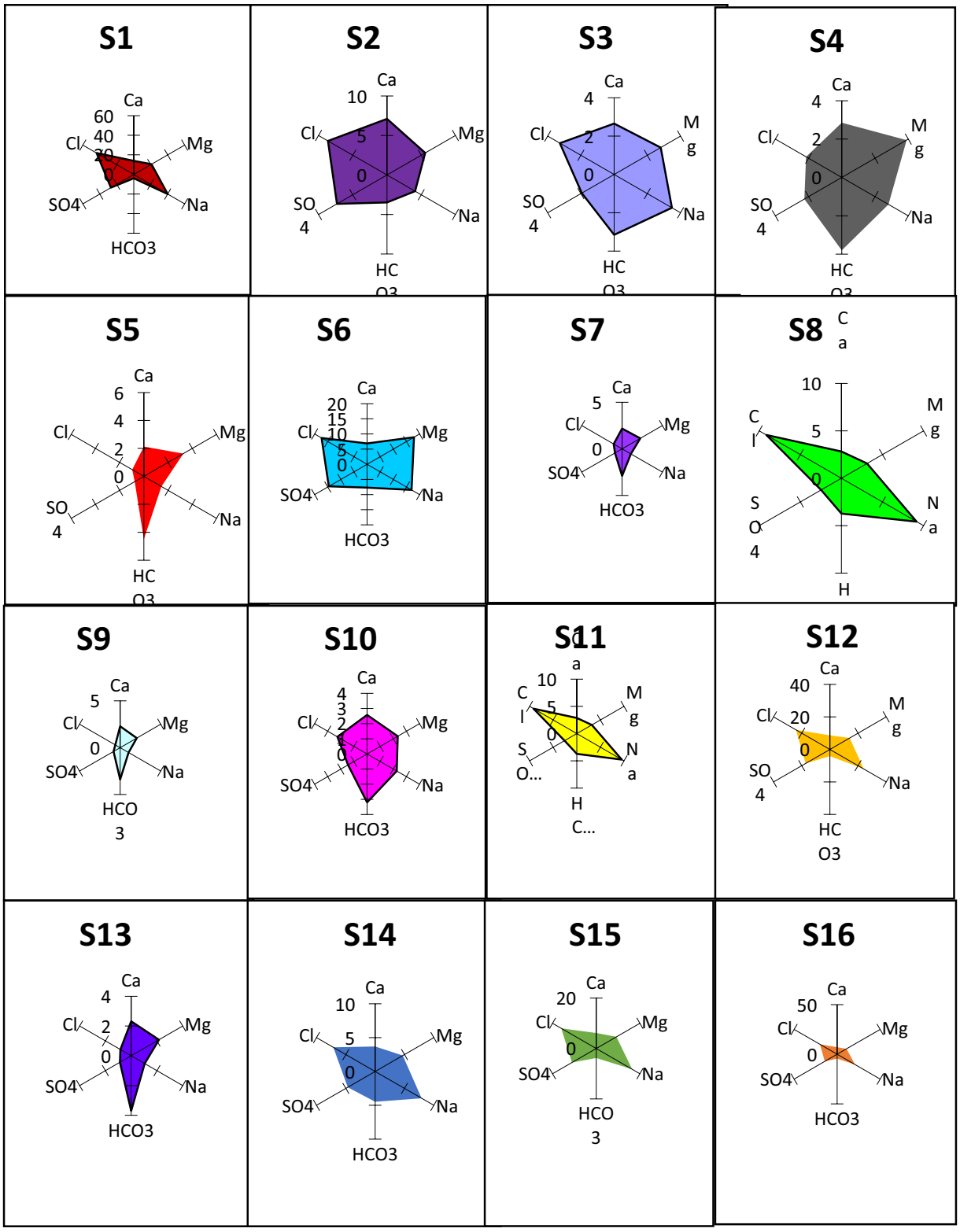

Fig. 6 Chemical analysis of the Elements in each sub-watershed

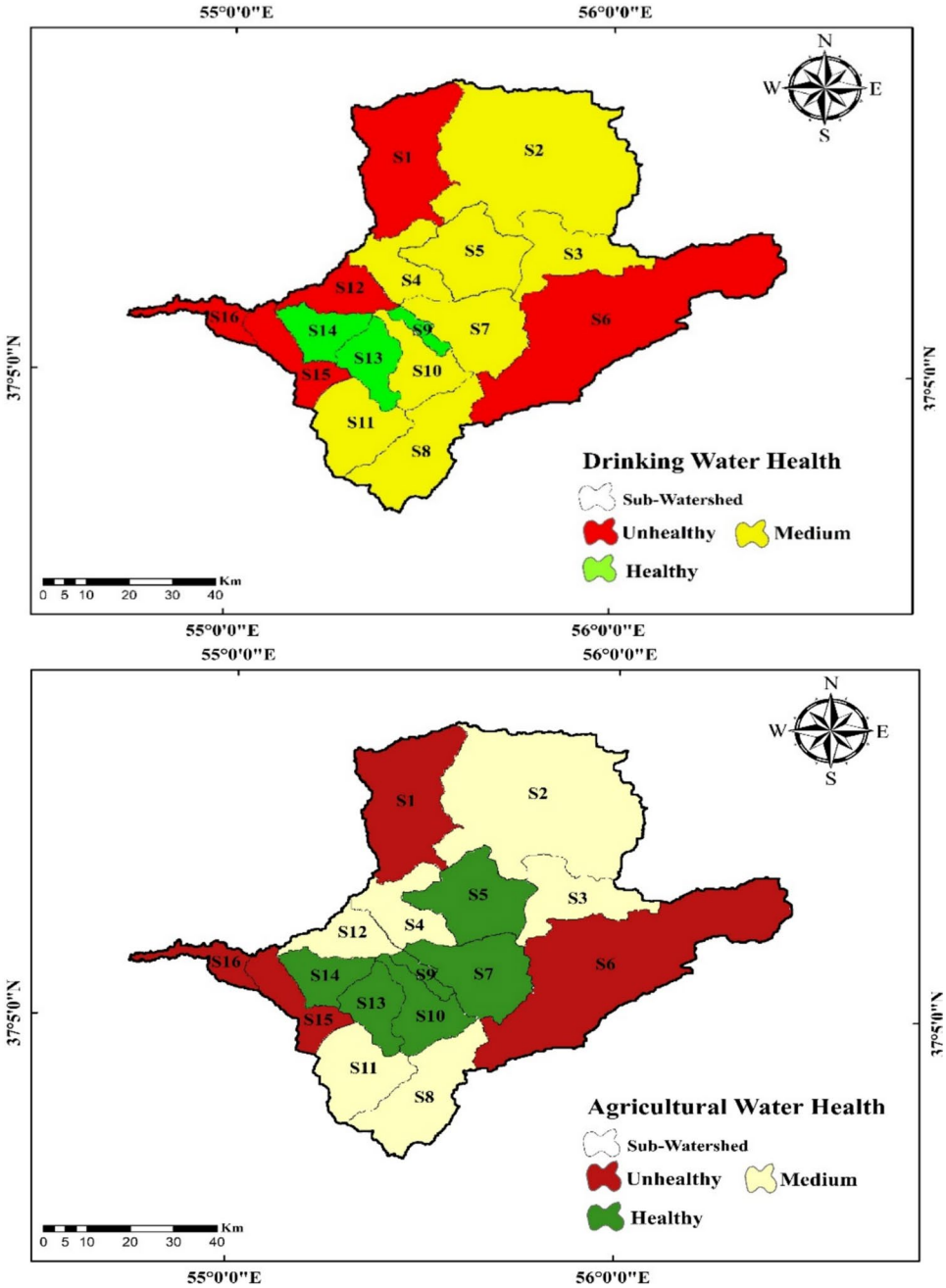

Fig.7 The health of drinking and agricultural water quality in each sub-watershed

exploitation, despite significant river accumulation. However, S16's resilience improves at 30- and 90-day timescales, supporting the findings of Wu et al. (2022) and Hazbavi et al. (2018a, b), which indicate that hydrological systems tend to stabilize over longer periods due to natural storage and recovery mechanisms.

Sub-watershed S14 exhibits the lowest 7-day minimum flow health, indicating significant vulnerability during dry periods. Likewise, S6, S9, and S12 consistently show low health values across all durations, reflecting poor drought resilience. Additionally, the disparity between 30- and 90-day minimum flow health in S12 and S15 points to ongoing difficulties in maintaining water accessibility during extended stress conditions.

Analysis of maximum flow health

Figure 4 illustrates that sub-watersheds S7, S3, and S9, characterized by forest cover, exhibit high permeability and demonstrate relative resilience to flood events. Conversely, sub-watershed S1 is highly vulnerable to flooding, attributed to inadequate vegetation cover, variable slopes, significant water erosion, and unregulated agricultural practices. Furthermore, sub-watersheds S6, S12, and S16, which have

limited vegetation cover, experience reduced permeability and an increased risk of flooding.

The sub-watershed S15 has a higher probability of flooding due to high runoff generation. Additionally, the sub-watersheds S8 and S11 require management to prevent flooding due to the density of the drainage network and the development of drainage systems. The sub-watersheds S10 and S4 have low resistance to flood flows, and sub-watershed S10 is particularly vulnerable.

Additionally, the sub-watershed S13, with the lowest health value (0.06), indicates significant vulnerability to flooding. The sub-watersheds S5 and S4, as two sub-watersheds located almost in the center of the watershed, fall into the unhealthy range in the health zoning and require strategic control and management. The sub-watershed S2, due to the high runoff volume, the diversity of various land uses, road construction, and the high runoff coefficient, has the potential to cause flooding; therefore, the health of the sub-watershed falls within the moderate range. Sub-watersheds with moderate resilience require a high level of management to improve their health. These findings of the current study are consistent with those of Shaabani Bazneshin et al. (2016), emphasizing the impact of factors such as location, vegetation cover, drainage density, land use, and permeability on flood potential.

Analysis of environmental flow health

Figure 5 illustrates that environmental flow health values range from 0.31 to 0.66, with S6 reporting the lowest health score and S15 the highest. Research indicates that the S15 sub-watershed has the highest discharge, creating optimal conditions for environmental flow and thereby supporting healthy ecosystems and biodiversity. Conversely, the low environmental flow health of S6 indicates potential stress on its environmental flow.

This sub-watershed suffers from a lack of vegetation cover and severe soil degradation, necessitating immediate action to restore or maintain ecological balance. Sub-watershed S16, located at the watershed outlet, has a value of 0.35, indicating a relatively unhealthy condition attributed to extensive agriculture and a high number of deep wells, which contribute to excessive water exploitation. Similarly, sub-watershed S14 has a value of 0.38, highlighting its unhealthy state due to overexploitation of water linked to diverse agricultural practices. These conditions raise concerns that require targeted intervention and management solutions.

The sub-watersheds S1, S2, S12, S13, S3, S4, S5, S8, S9, S10, and S11 are classified as having moderate health values according to the zoning map. This classification signifies that these areas offer suitable conditions for environmental flow. They maintain adequate water flow to support ecosystems, wildlife, and human needs, establishing them as benchmarks for other watersheds.

These sub-watersheds necessitate monitoring and preventive measures to prevent further degradation. The findings of this research are consistent with previous studies (Pakoksung et al. 2025; Zhao et al. 2025) regarding the impacts of land use changes in agricultural and urban areas on water quality and environmental flow. This is especially critical during the rainy season when runoff from agricultural areas significantly affects water quality. Furthermore, the importance of forested areas in stabilizing environmental conditions is emphasized.

Analysis of water quality health

Drinking water quality demonstrates significant variation across the sub-watersheds, with values ranging from 0.28 to 0.88. The highest scores were recorded in S13 (0.88) and S14 (0.72), indicating excellent water quality. Conversely, S1 (0.28), S6 (0.37), S15 (0.32), and S16 (0.37) exhibited the lowest scores, reflecting poor drinking water conditions. This discrepancy may be attributed to extensive agricultural activities and the application of chemical pesticides in these sub-watersheds, which contribute to increased concentrations of dissolved salts and contaminants.

The remaining sub-watersheds demonstrated moderate water quality, with values ranging from 0.34 to 0.69. Regarding agricultural water quality, sub-watersheds S1 (0.20), S6 (0.33), S12 (0.36), S15 (0.34), and S16 (0.36) were identified as having critically low health indices. These adverse conditions are closely associated with elevated electrical conductivity (EC > 1000 µS/cm in S1, S6, S15, and S16), heightened sodium adsorption ratio (SAR > 10 in S1, S12, and S16), and increased total hardness (TH).

These parameters typically decline as one approaches the watershed outlet, demonstrating the cumulative effects of salts, dissolved ions, and runoff from excessive fertilizer application. Increased electrical conductivity (EC) can impede water uptake by plants, hinder their growth, and ultimately result in plant mortality. Likewise, elevated water hardness, due to higher concentrations of calcium and magnesium, can lead to scale buildup in irrigation systems and diminish nutrient availability for crops.

These findings corroborate earlier research by Li et al. (2023) and Evans et al. (2019), which emphasized the negative impact of agricultural runoff, particularly from fertilizers and pesticides.

Sub-watersheds S7, S11, and S13 demonstrate excellent forest cover and exhibit more favorable conditions for both water quantity and quality. In contrast, sub-watersheds S2, S3, S4, S5, S8, and S10 are categorized within a moderate range concerning drinking water quality. This indicates that, with appropriate management strategies, their current

conditions can be maintained or improved. However, sub-watersheds with poor water quality require thorough assessments of pollution sources, along with the implementation of targeted conservation and water protection measures.

The results presented in Fig. 7 show that agricultural water health scores range from 0.19 to 0.76. The highest scores were found at S10 (0.72), S7 (0.73), S14 (0.75), and S13 (0.75), indicating good water quality for agricultural use. In contrast, S1 (0.20), S6 (0.24), and S15 (0.25) had the lowest scores, demonstrating poor suitability for irrigation. These findings are consistent with Zare Karizi et al. (2024), who reported that human activities, such as intensive farming and land tillage, significantly contribute to agricultural water pollution.

The remaining sub-watersheds, with scores ranging from 0.32 to 0.76, exhibit moderate water quality. This is influenced by various factors, including land use patterns, pollution sources, and the extent of water treatment or management interventions. Figure 8 presents the health index values for the criteria assessed in each sub-watershed.

Analysis of the health index

The health index was determined by calculating the weighted average of drinking water quality ($RRV_{_water\ drinking}$) and agricultural water quality ($RRV\text{-}_{Agricultural\ water}$). The selection of these two criteria is primarily due to their direct influence on human health and the environment. Recognizing the equal significance of both factors, the weights (W1 and W2) were assigned a value of 0.5. The calculation of the health index produced results, which are detailed in Table 7 for each sub-watershed. Health index values vary between 0 and 1. The closer the health index is to 1, the better the health of the watershed.

The correlation coefficient between health indicators based on each variable and the ordinal health index of water quality

The Pearson correlation coefficient between the variables and the health index was calculated based on the results presented in Table 8. Notably, two relationships approached the significance threshold: 90-day flow ($r = 0.45$, $p = 0.080$) and environmental flow ($r = 0.42$, $p = 0.105$). Although classical statistical significance ($p < 0.05$) was not achieved, these findings hold practical significance and may inform model development in future studies.

Table 9 illustrates a strong correlation among several indices, including RRV7, RRV30, and RRV90, with a correlation coefficient of $r = 0.876$. Additionally, a significant correlation exists between the RRV_Drinking Water index and the Agricultural Water Quality index, resulting in a

Fig. 8 The health index values for minimum flow, maximum flow, environmental flow, and water quality in each sub-watershed

Table 7 Calculation of the health index for each sub-watershed

Sub-watershed	RRV_water drinking	RRV-Agricultural water	Health index
S1	0.28	0.19	0.235
S2	0.59	0.66	0.625
S3	0.53	0.68	0.605
S4	0.37	0.24	0.305
S5	0.54	0.62	0.58
S6	0.64	0.72	0.68
S7	0.69	0.75	0.72
S8	0.45	0.73	0.59
S9	0.52	0.76	0.64
S10	0.56	0.66	0.61
S11	0.34	0.34	0.34
S12	0.72	0.75	0.735
S13	0.88	0.75	0.815
S14	0.52	0.67	0.595
S15	0.32	0.25	0.285
S16	0.37	0.32	0.345

Table 8 The correlation coefficient between health indicators based on each variable and the ordinal health index of water quality

Variables	Correlation coefficient (r)	*p*-Value (2-tailed)
RRV-Environmental flow	0.42 ^{ns}	0.10
RRV_Maximum flow	−0.28 ^{ns}	0.29
RRV Minimum flow 7 days	0.2 ^{ns}	0.45
RRV Minimum flow 30 days	0.35 ^{ns}	0.18
RRV Minimum flow 90 days	0.45 ^{ns}	0.080

^{ns}no significant in 0.01 and 0.05 significance levels

correlation coefficient of r = 0.953. This suggests a potential risk of multicollinearity if these indices are included in the model simultaneously. To evaluate collinearity among the indices, the Variance Inflation Factor (VIF) test was performed. A VIF value below 5 typically indicates that severe multicollinearity is not present among the variables.

Table 10 demonstrates that the indices for "environmental flow" and "maximum flow" exhibit VIF values of 1.32 and 1.25, respectively, signifying negligible collinearity with other variables. While certain variables, such as drinking water and agriculture, display structural correlation, the findings indicate that these indices uphold acceptable statistical independence. Consequently, they can be incorporated into the analytical model without concerns regarding multicollinearity.

The statistical interpretation of "lack of alignment" is substantiated by evidence indicating that these indices exhibit low dependency on other variables. Furthermore, Table 11 displays the results of an ANOVA test performed to assess differences among the RRV indices. Although the F statistic was notably high (F = 19.23), the significance level (Sig. = 0.177) suggests that the differences in mean indices are not statistically significant ($p > 0.05$).

Analysis of the calculated criteria at the watershed level

The results illustrated in Fig. 9 detail the calculations for minimum flows over 7, 30, and 90 days, as well as maximum flow, environmental flow, and water quality at the watershed level.

The health index of the Gorganroud watershed, determined by the 7-day minimum flow, is 0.16. This classification indicates that the watershed is considered very unhealthy, rendering it susceptible to water scarcity during short-term periods. As a result, water accessibility is inadequate, posing a significant risk to the ecosystems within the watershed.

The minimum flow over 30 days in the watershed area is recorded at 0.32, indicating long-term water stress. In contrast, a healthy flow sustained for at least 90 days at the watershed level is documented at 0.45, suggesting stable water availability during that timeframe. Furthermore, the maximum flow in the Gorganrood watershed is assessed at 0.34, reflecting an unhealthy condition associated with flooding. These findings are consistent with those of Tavosi et al. (2025), underscoring the necessity for management measures at both the watershed level and within all sensitive sub-watersheds.

The environmental flow health of the Gorganroud watershed is rated at 0.47. This suggests that the watershed has a limited capacity to meet the needs of humans, animals, and plants. Without strategic management, it is likely to follow an unsustainable trend. In contrast to the Shazand watershed, which Sadeghi et al. (2019a, b) identified as having no sub-watersheds in a healthy condition (IWHI < 0.3), the Gorganroud watershed exhibits moderate health, with an IWHI of 0.47 for environmental flows, indicating greater resilience.

Chamani et al. (2023) reported a significant enhancement in the water health index for the Atrak watershed. Their findings suggest that rainfall is anticipated to increase by 31.74% under the RCP4.5 scenario and by 29.79% under the RCP8.5 scenario. Furthermore, the metrics for Rel, Res, and Vul exhibited minimal changes in future scenarios compared to the baseline period. However, the overall watershed health index declined from 0.46 during the baseline period to 0.42 under RCP4.5 and to 0.43 under RCP8.5.

These findings can serve as a foundation for adaptive planning in water resource management. Furthermore, a comparison with UNESCO's water resource resilience

Table 9 Pearson correlation analysis between research indices

		RRV Minimum flow 7 days	RRV Minimum flow 30 days	RRV Minimum flow 90 days	RRV-Environmental flow	RRV_ Maximum flow	RRV_Drinking water quality	RRV-Agricultural water quality
RRV Minimum flow 7 days	Pearson correlation	1	0.680**	0.522*	−0.088	0.104	−0.016	−0.004
	Sig. (2-tailed)		0.004	0.038	0.745	0.701	0.952	0.989
	N	16	16	16	16	16	16	16
RRV Minimum flow 30 days	Pearson correlation	0.680**	1		−0.069	0.044	0.142	0.055
	Sig. (2-tailed)	0.004		0.000	0.798	0.870	0.601	0.839
	N	16	16	16	16	16	16	16
RRV Minimum flow 90 days	Pearson correlation	0.522*	0.876**	1	0.035	−0.123	0.403	0.302
	Sig. (2-tailed)	0.038	0.000		0.899	0.651	0.122	0.255
	N	16	16	16	16	16	16	16
RRV-Environmental flow	Pearson correlation	−0.088	−.069	0.035	1	−0.339	−0.164	−0.145
	Sig. (2-tailed)	0.745	0.798	0.899		0.199	0.543	0.592
	N	16	16	16	16	16	16	16
RRV_Maximum flow	Pearson correlation	0.104	0.044	−0.123	−0.339	1	−0.159	−0.131
	Sig. (2-tailed)	0.701	0.870	0.651	0.199		0.557	0.629
	N	16	16	16	16	16	16	16
RRV_ drinking water quality	Pearson Correlation	−0.016	0.142	0.403	−0.164	−0.159	1	0.953**
	Sig. (2-tailed)	0.952	0.601	0.122	0.543	0.557		.000
	N	16	16	16	16	16	16	16
RRV-Agricultural water quality	Pearson Correlation	−0.004	0.055	0.302	−0.145	−0.131	0.953**	1
	Sig. (2-tailed)	0.989	0.839	0.255	0.592	0.629	0.000	
	N	16	16	16	16	16	16	16

Table 10 Calculation of the variance inflation factor (VIF)

Index	VIF
RRV-Environmental flow	1.32
RRV_Maximum flow	1.25
RRV Minimum flow 7 days	2.16
RRV Minimum flow 30 days	8.31
RRV Minimum flow 90 days	7.97
RRV_water drinking quality	15.43
RRV-Agricultural water quality	13.39

indicators—where a resilience threshold greater than 0.5 is deemed sustainable—indicates that sub-watersheds S15 and S8 meet these criteria, while sub-watersheds S6 and S16 necessitate immediate management interventions.

The health of drinking water and agriculture at the watershed level is generally favorable, with scores of 0.52 and 0.57 under optimal conditions. However, population growth and industrialization are contributing to an increased influx of pollutants from industrial activities into water sources. To ensure a safe supply of water for drinking and agricultural purposes in the Gorganroud watershed, a comprehensive strategy is required to safeguard the quality and integrity of water resources across all dimensions.

Conclusion and recommendation

This study evaluated the resilience, reliability, and vulnerability of water resources within the Gorganroud watershed, taking into account the impacts of minimum and maximum flows, environmental flow, and water quality. The results revealed that the health indices for minimum flow, assessed over 7-, 30-, and 90-day periods, were 0.16, 0.32, and 0.45, respectively. These findings indicate that short-term minimum flows are particularly vulnerable, underscoring the necessity for enhanced management and regulation of water resources in the watershed.

Table 11 Calculation analysis of variance of health indicators in the Gorganroud watershed

ANOVA						
		Sum of Squares	df	Mean Square	F	Sig
RRV-Environmental flow	Between Groups	0.121	14	0.009	19.230	0.177
	Within Groups	0.000	1	0.000		
	Total	0.122	15			
RRV_Maximum flow	Between Groups	0.287	14	0.020	2.426	0.469
	Within Groups	0.008	1	0.008		
	Total	0.295	15			
RRV Minimum flow 7 days	Between Groups	0.072	14	0.005	–	–
	Within Groups	0.000	1	0.000		
	Total	0.072	15			
RRV Minimum flow 30 days	Between Groups	0.709	14	0.051	40.503	0.123
	Within Groups	0.001	1	0.001		
	Total	0.710	15			
RRV Minimum flow 90 days	Between Groups	0.515	14	0.037	1.178	0.628
	Within Groups	0.031	1	0.031		
	Total	0.547	15			
RRV_water drinking quality	Between Groups	0.400	14	0.029	571.071	0.033
	Within Groups	0.000	1	0.000		
	Total	0.400	15			
RRV-Agricultural water quality	Between Groups	0.362	14	0.026	57.396	0.103
	Within Groups	0.000	1	0.000		
	Total	0.362	15			

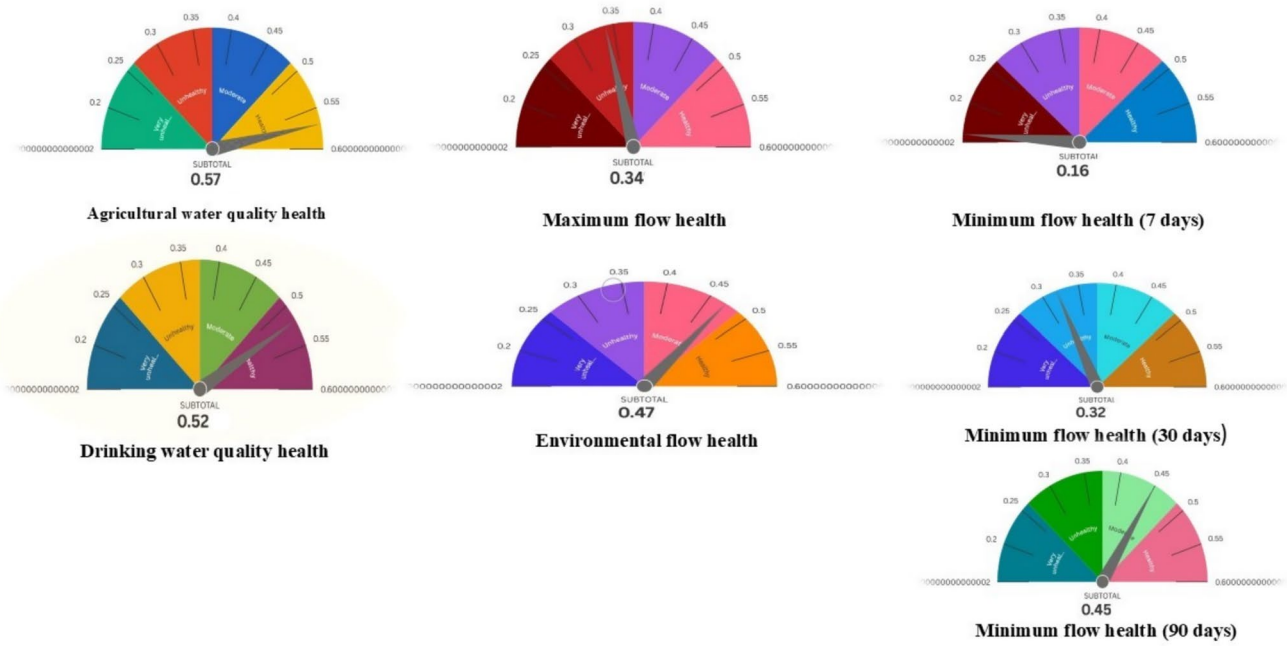

Fig. 9 Analysis of the calculated criteria at the watershed level

The assessment of maximum flow conditions indicated a significant concern within the watershed, particularly in the S13 sub-watershed, which exhibited the highest level of vulnerability. This underscores the urgent necessity for flood risk management and the implementation of protective measures to mitigate potential damage.

The water quality results show that the drinking water quality index for sub-watersheds S1, S6, S12, S15, and S16 is classified as "very unhealthy." This decline is mainly due to high chloride concentrations, excessive use of chemical fertilizers in agriculture, industrial discharges, and the presence of suspended solids like sand and silt. While the overall watershed-level health index for drinking water quality is 0.52, which is considered relatively healthy, specific issues within these sub-watersheds reveal significant management challenges.

In the context of agricultural water quality, the primary parameters identified were electrical conductivity (EC), total hardness (TH), and pH. Significant spatial variability was observed, with TH and pH values exhibiting notable fluctuations from upstream to downstream. Specifically, sub-watersheds S1, S6, S15, and S16 indicated unfavorable conditions for agricultural water use.

The health of environmental flows is considerably affected by human activities, leading to an overall watershed index of 0.47, which reflects a "moderate" condition in accordance with watershed health zoning criteria. In light of the compounded effects of climate change and recurring droughts, the absence of strategic management and protective policies poses a significant threat to the reliability of water resources and the long-term requirements of both human populations and ecological communities within the watershed.

This research utilizes the innovative RRV multi-criteria approach specifically designed for the Gorganroud watershed, thereby significantly advancing our understanding of watershed dynamics. It offers valuable insights for sustainable water resource management in similarly complex systems. The findings can serve as a practical roadmap for managers and policymakers, enabling the implementation of effective management strategies through the integrated management of critical and priority sub-watersheds.

Recommended strategies involve implementing advanced monitoring systems to track water flow and quality in vulnerable sub-watersheds, such as S14 and S16. Additionally, enforcing stricter controls to reduce agricultural runoff in areas like S1, S6, and S15 is essential. Promoting local community engagement in sustainable agricultural practices will further enhance watershed resilience. Together, these measures aim to prevent watershed degradation and address emerging threats, particularly in the water resources sector.

The RRV approach aligns with Iran's national water security framework and supports policies like the national water management program by identifying priority sub-watersheds, such as S6 and S16, for targeted intervention. However, its implementation may face challenges, including legal restrictions on land use changes and conflicts between agricultural and conservation priorities. Addressing these issues will require collaborative efforts among stakeholders and legal reforms.

Smart monitoring systems are utilized to measure minimum and maximum flow levels in vulnerable sub-watersheds (S14, S16, S6, S15) in real time. Additionally, these systems can monitor electrical conductivity (EC) and total hardness (TH) data, offering early warnings for potential droughts or floods.

Small reservoirs in the S14 and S16 sub-watersheds could greatly increase water storage capacity. A preliminary cost–benefit analysis indicates that constructing each reservoir, with an estimated cost of $1 million to $2 million, could provide annual flood reduction benefits valued at $3 million.

It is recommended to immediately deploy smart and cost-effective quality monitoring systems in the S14 and S16 sub-watersheds. Additionally, implementing long-term afforestation programs in the S1 and S6 sub-watersheds is essential for mitigating the gradual degradation of water quality.

Future studies should utilize climate models, such as CMIP6 outputs, to evaluate the potential impacts of climate change on the ecological health of the watershed.

The green engineering strategy for flood mitigation in watersheds focuses on nature-based solutions and sustainable design to reduce environmental impacts while improving runoff and water flow management. This approach, grounded in principles such as ecosystem preservation and resource efficiency, is particularly crucial in areas like S2, where road construction has led to increased runoff. The sustainable watershed management framework, adopted by organizations like the EPA and UNEP, serves as a guiding model. A notable example is a project in Portland, Oregon, which achieved a 70% reduction in urban runoff by incorporating green infrastructure elements such as rain gardens and permeable surfaces.

To enhance long-term reliability, we recommend conducting a comprehensive evaluation of the Gorganroud watershed's health every 5 to 10 years. This assessment should integrate updated hydrological and water quality data to effectively address the impacts of climate change and evolving land use patterns.

To engage stakeholders, especially farmers in the S1, S6, S15, and S16 sub-watersheds, we should organize workshops and offer incentives that promote the adoption of sustainable practices. These practices should focus on reducing pesticide usage and implementing cover cropping.

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Data availability Data will be made available on request.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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